Irreducibility in $\mathbb{K}[x]$

- **Definition 1**
  A polynomial $p(x) \in \mathbb{K}[x]$ with $\deg(p(x)) > 0$ is **irreducible** over $\mathbb{K}$ if whenever $p(x) = q(x)h(x)$ for some $q(x), h(x) \in \mathbb{K}[x]$ then either $q(x)$ or $h(x)$ is a constant polynomial. Otherwise, $p(x)$ is **reducible**.
  - **Example 1.1**
    The polynomial $p(x) = x^2 - 3$ is irreducible over $\mathbb{Q}$ but is reducible over $\mathbb{R}$ since
    $$p(x) = \left(x + \sqrt{3}\right)\left(x - \sqrt{3}\right).$$

- **Definition 2**
  Let $p(x) \in \mathbb{K}[x]$. An element $\alpha \in \mathbb{K}$ is a root of $p(x)$ if $p(\alpha) = 0$.
  - **Example 2.1**
    The polynomial $p(x) = (x-1)(x-2)(x+3)$ has roots $1, 2$ and -3.
    The polynomial $p(x) = x^2 + 1$ has no roots in $\mathbb{R}$ but has two roots i and $-\mathrm{i}$ in $\mathbb{C}$.

- **Definition 3**
  Let $p(x), q(x) \in \mathbb{K}[x]$. We say that $q(x)$ divides $p(x)$, denoted $q(x) \mid p(x)$, if there exists $h(x) \in \mathbb{K}[x]$ such that
  $$p(x) = q(x)h(x).$$
  - **Theorem 3.1 (Factor Theorem)**
    Let $p(x) \in \mathbb{K}[x]$ and $\alpha \in \mathbb{K}$, then α is a root of $p(x)$ if and only if $(x - \alpha) \mid p(x)$.
    - **Proof 3.1.1**
      Suppose $p(x) \in \mathbb{K}[x]$ and $\alpha \in \mathbb{K}$.
      ($\Longrightarrow$) Suppose that α is a root of $p(x)$ then $p(\alpha) = 0$ and by the remainder theorem there exists $q(x) \in \mathbb{K}[x]$ such that
      $$p(x) = (x-\alpha)q(x) + p(\alpha) = (x-\alpha)q(x).$$
      That is, $(x - \alpha) \mid p(x)$.

      ($\Longleftarrow$) Suppose that $(x-\alpha) \mid p(x)$. By definition, there exists $q(x) \in \mathbb{K}[x]$ such that $p(x) = (x-\alpha)q(x)$. Hence $p(\alpha) = 0$ and thus α is a root of $p(x)$.
  - **Example 3.2**
    Consider $p(x) = x^3 - 2x^2 - 5x + 6$. Since $p(1) = 1 - 2 - 5 + 6 = 0$ then by the factor theorem $(x-1) \mid p(x)$. Furthermore
    $$p(x) = (x-1)\left(x^2 - x - 6\right).$$
  - **Corollary 3.3**
    Let $p(x) \in \mathbb{K}[x]$ with $\deg(p(x)) > 1$. If $p(x)$ has a root in $\mathbb{K}$ then $p(x)$ is reducible over $\mathbb{K}$.
    - **Proof 3.3.1**
      Let α be a root of $p(x)$, then by the factor theorem $(x-\alpha) \mid p(x)$, that is, there exists a polynomial $q(x) \in \mathbb{K}[x]$ such that $p(x) = (x-\alpha)q(x)$. Since
      $$\deg(p(x)) = \deg((x-\alpha)q(x)) = 1 + \deg(q(x)) > 1.$$
      Then $q(x)$ is not a constant polynomial and thus $p(x)$ is reducible over $\mathbb{K}$.
  - **Remark 3.4**

The converse is false. Consider $p(x)=x^4+2x^2+1\in\mathbb{R}[x]$, then $p(x)$ is reducible over $\mathbb{R}$ since

$$p(x)=\left(x^2+1\right)\left(x^2+1\right).$$

But $p(x)$ has no roots in $\mathbb{R}$.

- **Proposition 3.5**
  Let $p(x)\in\mathbb{K}[x]$.

  1. If $\deg(p(x))=1$, then $p(x)$ is irreducible over $\mathbb{K}$.
  2. If $\deg(p(x))$ is 2 or 3, then $p(x)$ is irreducible over $\mathbb{K}$ if and only if $p(x)$ has no roots in $\mathbb{K}$.

  - **Proof 3.5.1**

    1. If $p(x)=q(x)h(x)$ then

       $$1=\deg(p(x))=\deg(q(x)h(x))=\deg(q(x))+\deg(h(x)).$$

       Hence $\deg(q(x))=0\vee\deg(h(x))=0$, that is, $q(x)$ is a constant polynomial or $h(x)$ is a constant polynomial. Therefore $p(x)$ is irreducible over $\mathbb{K}$.
    2. ($\Longrightarrow$) Suppose that $p(x)$ is irreducible over $\mathbb{K}$ then by the corollary $p(x)$ has no roots in $\mathbb{K}$.

       ($\Longleftarrow$) Suppose that $p(x)$ has no roots in $\mathbb{K}$. If $p(x)$ is reducible over $\mathbb{K}$ then $p(x)=q(x)h(x)$ for some nonconstant polynomials $q(x),h(x)\in\mathbb{K}[x]$. Hence

       $$2=\deg(p(x))=\deg(q(x)h(x))=\deg(q(x))+\deg(h(x)).$$

       Then $\deg(q(x))=\deg(h(x))=1$, that is, $q(x)=ax+b$ with $a,b\in\mathbb{K}$ and $a\neq 0$ with $q(-a^{-1}b)=0$ and $-a^{-1}b\in\mathbb{K}$. Therefore $-a^{-1}b$ is a root of $p(x)$ which is a contradiction.
       Similarly if $\deg(p(x))=3$. Thus $p(x)$ is irreducible over $\mathbb{K}$.

- **Theorem 3.6**
  Let $p(x)\in\mathbb{K}[x]$ with $p(x)\neq 0$. If $\deg(p(x))=n$ then $p(x)$ has at most n roots in $\mathbb{K}$.

  - **Proof 3.6.1**
    By induction on n.

    1. **Base step** If $n=1$ then $p(x)=ax+b$ with $a,b\in\mathbb{K}$ and $a\neq 0$ which has exactly one root.
    2. **Inductive step** Suppose that the result is true for n and consider $\deg(p(x))=n+1$. If $p(x)$ has no roots in $\mathbb{K}$ there is nothing to prove. If $p(x)$ has roots in $\mathbb{K}$, let $\alpha\in\mathbb{K}$ be a root of $p(x)$. Then by the factor theorem $(x-\alpha)\mid p(x)$, that is,

       $$p(x)=(x-\alpha)q(x)\quad\text{for some } q(x)\in\mathbb{K}[x].$$

       Furthermore

       $$n+1=\deg(p(x))=\deg((x-\alpha)q(x))=1+\deg(q(x)).$$

       Then $\deg(q(x))=n$ and by the inductive hypothesis $q(x)$ has at most n roots in $\mathbb{K}$. If $\beta\in\mathbb{K}$ is another root of $p(x)$ then $p(\beta)=(\beta-\alpha)q(\beta)=0$. So we obtain that $\beta=\alpha$ or $q(\beta)=0$. Therefore β is among the at most n roots, that is $p(x)$ has at most $n+1$ roots.

- **Definition 4**
Let $p(x) \in \mathbb{K}[x]$ with $p(x) \neq 0$ and let $\alpha \in \mathbb{K}$ be a root of $p(x)$. The **multiplicity** of α is the largest integer m such that

$$(x-\alpha)^m \mid p(x).$$

  - **Remark 4.1**
  Equivalently, α has multiplicity m when

  $$p(x) = (x-\alpha)^m q(x)$$

  with $q(\alpha) \neq 0$.
  - **Example 4.2**
  If $p(x) = (x-2)^3(x+1)$ then 2 is a root of $p(x)$ with multiplicity 3 and -1 is a root of $p(x)$ with multiplicity 1.
  - **Theorem 4.3 (Fundamental theorem of algebra)**
  If $p(x) \in \mathbb{C}[x]$ with $\deg(p(x)) \geqslant 1$ then $p(x)$ has a root in $\mathbb{C}$. Equivalently, every polynomial $p(x) \in \mathbb{C}[x]$ has exactly n roots in $\mathbb{C}$ where roots are counted with multiplicities.
  - **Corollary 4.4**
  A nonconstant polynomial in $\mathbb{C}[x]$ is irreducible if and only if it has degree 1.
  - **Corollary 4.5**
  Let $p(x) \in \mathbb{C}[x]$ with $\deg(p(x)) = n \geqslant 1$. Then

  $$p(x) = \alpha_0 (x-\alpha_1)^{m_1} (x-\alpha_2)^{m_2} \cdots (x-\alpha_s)^{m_s}$$

  where $\alpha_1, \alpha_2, \cdots, \alpha_s$ are the distinct roots of $p(x)$, $m_1, m_2, \cdots, m_s$ are the multiplicities with $m_1 + m_2 + \cdots + m_s = n$.
  - **Theorem 4.6**
  Let $p(x) \in \mathbb{R}[x]$. If $z = a + b\,\mathrm{i}$ is a root of $p(x)$ then $\overline{z} = a - b\,\mathrm{i}$ is also a root of $p(x)$.
    - **Proof 4.6.1**
    Let $p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \in \mathbb{R}[x]$ and let $z = a + b\,\mathrm{i}$ be a root of $p(x)$, that is, $p(z) = 0$. Hence

    $$\begin{aligned} p\left(\overline{z}\right) &= a_0 + a_1 \overline{z} + a_2 (\overline{z})^2 + \cdots + a_n (\overline{z})^n \\ &= a_0 + a_1 \overline{z} + a_2 \overline{z}^2 + \cdots + a_n \overline{z}^n \\ &= \overline{a_0} + \overline{a_1 z} + \overline{a_2 z^2} + \cdots + \overline{a_n z^n} \\ &= \overline{a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n} \\ &= \overline{p(z)} = \overline{0} = 0. \end{aligned}$$

  - **Corollary 4.7**
  Every polynomial $p(x) \in \mathbb{R}[x]$ with odd degree has at least one real root.
    - **Proof 4.7.1**
    Suppose that $\deg(p(x)) = 2k+1$ for some $k \in \mathbb{N}_0$ then $p(x)$ has roots $\alpha_1, \alpha_2, \cdots, \alpha_{2k+1} \in \mathbb{C}$ and by Theorem 4.6, $\overline{\alpha_1}, \overline{\alpha_2}, \cdots, \overline{\alpha_{2k+1}}$ are also roots of $p(x)$. There exist some roots such that $\alpha_i = \overline{\alpha_i}$ and thus $\alpha_i \in \mathbb{R}$.
  - **Theorem 4.8 (Rational roots theorem)**
  Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{Z}[x]$ with $a_0, a_n \neq 0$ and let $\alpha = \dfrac{r}{s} \in \mathbb{Q}$ with $r \in \mathbb{N}, s \in \mathbb{Z}$ and r, s are coprime. If α is a root of $p(x)$ then we have that $r \mid a_0$ and $s \mid a_n$. For example, let

  $$p(x) = 3x^3 + 2x^2 - 2x - 8.$$

  The possible rational roots are

$$\pm 1, \pm 2, \pm 4, \pm 8, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}, \pm\frac{8}{3}.$$

- **Proof 4.8.1**

  Suppose $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{Z}[x]$ with $a_0, a_n \neq 0$. Suppose $p\left(\frac{r}{s}\right) = 0$ for some coprime $r, s \in \mathbb{Z}$, that is

  $$p\left(\frac{r}{s}\right) = a_0 + a_1\left(\frac{r}{s}\right) + \cdots + a_n\left(\frac{r}{s}\right)^n = 0.$$

  To clear denominators, multiplying both sides by s^n yields

  $$a_0 s^n + a_1 r s^{n-1} + \cdots + a_{n-1} r^{n-1} s + a_n r^n = 0.$$

  Shifting the $a_0 s^n$ term to the right side and factoring out r on the left side produces

  $$r\left(a_1 s^{n-1} + a_2 r s^{n-2} + \cdots + a_{n-1} r^{n-2} s + a_n r^{n-1}\right) = -a_0 s^n.$$

  Thus, r divides $a_0 s^n$ But r is coprime to s and therefore to s^n, so by Euclid's lemma r must divide the remaining factor a_0.

  $$r \mid a_0.$$

  On the other hand, shifting the $a_n r^n$ term to the right side and factoring out s on the left side produces

  $$s\left(a_0 s^{n-1} + a_1 r s^{n-2} + \cdots + a_{n-1} r^{n-1}\right) = -a_n r^n.$$

  Similarly it follows that s divides a_n. Therefore, the rational roots theorem is proved.